

Seagrass Meadows as Localised pH Refugia Against Ocean Acidification

A Project Report on Data Sources, Methodology, and Analytical Framework

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Abstract

Ocean acidification (OA), driven by rising atmospheric CO_2 , is lowering seawater pH and aragonite saturation state (Ω_{ar}) globally, threatening calcifying marine organisms. Seagrass meadows can modify local carbonate chemistry through photosynthetic CO_2 uptake and other biogeochemical processes, resulting in changes in DIC, TA, pH and Ω_{ar} (Duarte et al., 2013; Cyronak et al., 2018). This project investigates these carbonate-system differences for seagrass banks in the Bahamas using a 10-m resolution global seagrass extent map (Peng et al., 2026), Copernicus Marine Service fields for total alkalinity (TA) and dissolved inorganic carbon (DIC), and SOCATv2023 for sea surface temperature (SST) and salinity (SSS), combined through PyCO2SYS to solve the local carbonate system. The workflow was developed progressively, beginning with Google Earth Engine (GEE) for spatial data handling, followed by QGIS-based dataset verification and spatial analysis, selection of suitable meadow/open-water point pairs, and Python-based extraction and carbonate-system calculations. GLODAP was evaluated as a possible DIC source but found to have no usable data coverage in the study area and was therefore excluded. This report documents the complete data-selection and analytical workflow and provides the framework for evaluating ΔpH and $\Delta\Omega_{ar}$ between seagrass-associated and open-water locations.

1 Introduction and Motivation

Since the pre-industrial era, the ocean has absorbed roughly a third of anthropogenic CO_2 emissions, lowering global mean surface pH and reducing carbonate ion availability for shell- and skeleton-building organisms. Coastal ecosystems, however, are chemically heterogeneous: submerged vegetation can alter local carbonate chemistry through photosynthesis, respiration, sediment interactions and hydrodynamic processes. Seagrass systems have consequently been investigated through measurements and models of DIC, TA, pH and related carbonate parameters (Duarte et al., 2013; Kowek et al., 2018; Cyronak et al., 2018).

The present project focuses on the Bahamas, where extensive shallow carbonate banks provide large areas of seagrass habitat adjacent to surrounding open water. Rather than attempting to reproduce a single existing experimental design, the project combines methods and datasets from different areas of marine science: satellite-based seagrass mapping, spatial GIS analysis, gridded oceanographic datasets, observational carbonate-system data and numerical carbonate chemistry calculations.

The objective is therefore to develop a reproducible workflow for identifying suitable seagrass and open-water locations and quantifying their carbonate-system differences using real environmental data.

2 Data Sources

Table 1 summarises the data products evaluated and their final role in the workflow. Dataset selection was performed iteratively rather than assuming that all available products would provide usable data over the study area.

Source	Variable(s) used	Coverage	Role / status
Peng et al. (2026), Zenodo	Seagrass extent, 10 m	Global, 2019–2020 subset	Defines the spatial extent of mapped seagrass meadows and provides the basis for selecting meadow locations.
Copernicus Marine Service	TA, DIC/TIC	Regional gridded product	Primary carbonate-system inputs used at the selected sampling locations.
SOCATv2023	SST, SSS	Global gridded product	Provides temperature and salinity associated with the selected sampling locations.
GLODAPv2.2023	DIC (intended)	Limited spatial coverage	Evaluated as an independent DIC source but excluded because suitable observations were unavailable within the study area.
PyCO2SYS	pH, Ω_{ar}	Computed	Used to calculate carbonate-system properties from DIC, TA, temperature and salinity.

Table 1: Data sources evaluated and their role in the final analysis.

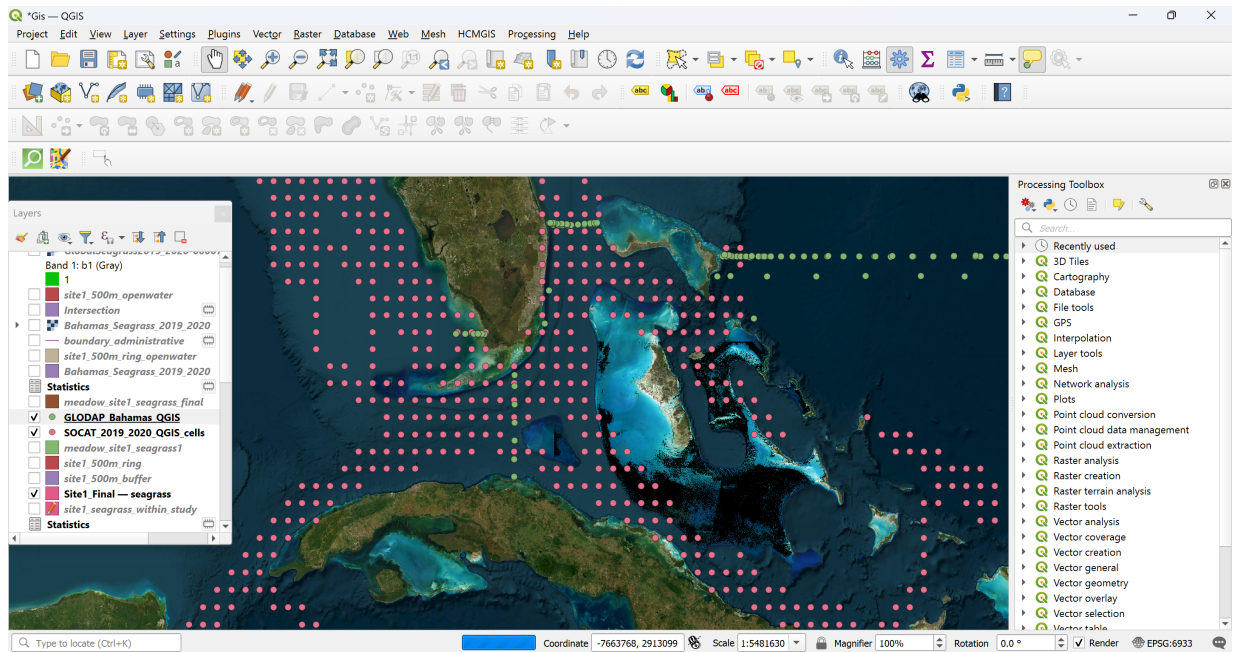


Figure 1: **Dataset Coverage and Study Area (QGIS).** Spatial distribution of the datasets evaluated for the Bahamas study region, including the mapped seagrass extent, SOCAT cells, GLODAP observations, and Copernicus carbonate-system grids.

Seagrass mapping dataset. The seagrass extent was obtained from the *GlobalSeagrass2019_2020* dataset, providing global seagrass mapping at approximately 10-m spatial resolution (Peng et al., 2026). The dataset was downloaded from Zenodo and subsequently clipped to the selected Bahamas study region.

Copernicus Marine Service. The Copernicus Marine carbonate-system product was used for total alkalinity and total/dissolved inorganic carbon. The product provides the carbonate variables on a substantially coarser grid than the 10-m seagrass map.

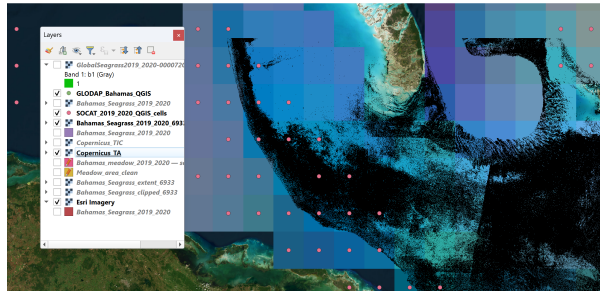
SOCATv2023. SOCATv2023 was used for sea surface temperature and salinity. These variables provide the physical conditions required alongside DIC and TA when solving the carbonate system.

GLODAP. GLODAPv2.2023 was initially considered as an independent DIC source. Its spatial coverage was checked directly in QGIS. Suitable observations were not available within the selected Bahamas sampling area, so the dataset was not used in the final calculation.

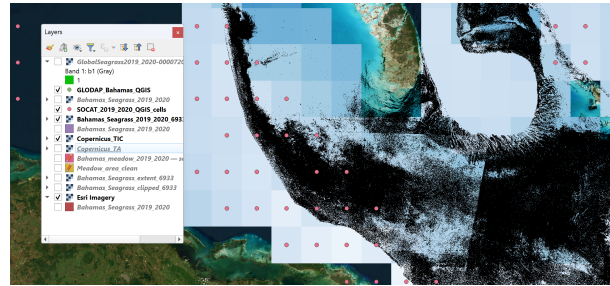
3 Study Area

The study area lies on the Great Bahama Bank and the Cay Sal Bank, Bahamas, spanning approximately 22.1–24.9°N and 76.6–78.9°W.

The region contains extensive shallow carbonate banks and large areas of submerged seagrass habitat. The combination of high-resolution seagrass mapping and available gridded carbonate-system data makes the region suitable for testing the proposed workflow.



(a) TA Copernicus



(b) TIC Copernicus

Figure 2: Final Study-Area Map: Spatial reference and carbonate-system data layers. (a) Total Alkalinity (TA) gridded product and (b) Total Inorganic Carbon (TIC) gridded product. These grids, along with SOCAT and seagrass extent, are used to evaluate local carbonate-system differences.

4 Methodology

4.1 Overall Workflow

The analysis was developed as a sequence of spatial and computational steps rather than as a single fixed procedure:

GEE → QGIS → Dataset Verification → Point Selection → Python Extraction → PyCO2SYS → ΔpH , $\Delta \Omega_{ar}$

The main stages are described below.

4.2 Initial Spatial Processing in Google Earth Engine

The initial approach was developed using Google Earth Engine (GEE) because of its ability to handle large satellite-derived spatial datasets and perform cloud-based raster processing. GEE was investigated as the

starting point for working with the global seagrass mapping data and identifying the relevant Bahamas region.

The use of GEE also follows established remote-sensing workflows in which Sentinel-2 imagery and cloud-based processing are used for large-scale seagrass mapping (Traganos et al., 2018). In the present project, however, a completed global seagrass map was ultimately used rather than independently training a new seagrass classifier.

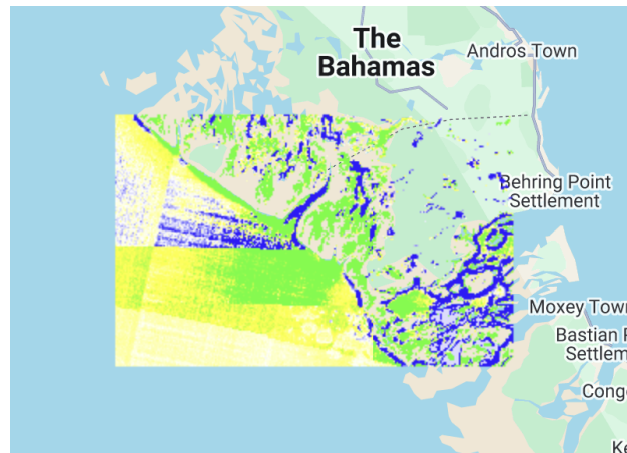


Figure 3: **Initial GEE Processing.** Screenshot showing the initial GEE-based study-area selection and seagrass raster processing.

4.3 Transition to QGIS

The workflow was subsequently transferred to QGIS for detailed spatial inspection and dataset comparison. QGIS allowed the different raster and vector products to be displayed simultaneously, making it possible to inspect whether each dataset actually covered the intended sampling region.

The seagrass extent was clipped to the Bahamas study area and spatially compared against the Copernicus and SOCAT grids. The GLODAP point distribution was also inspected at this stage.

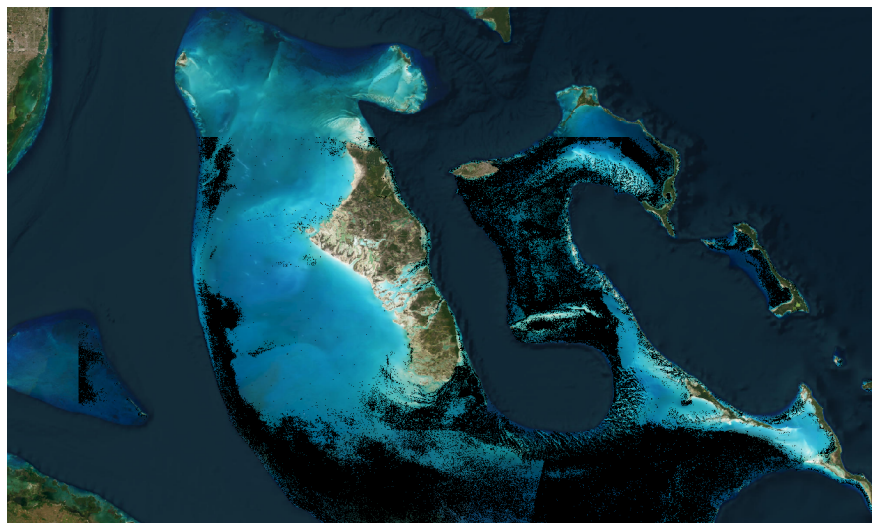


Figure 4: **Seagrass Extent after QGIS Clipping.** The seagrass map after clipping the 10-m GlobalSeagrass2019–2020 dataset to the Bahamas study region.

4.4 Dataset Testing and Verification

Before extracting values for the final analysis, each candidate dataset was tested for spatial coverage and compatibility with the study area.

The seagrass map was checked to ensure that the selected locations were actually within mapped meadow areas. Copernicus TA and DIC/TIC layers were checked for valid grid cells over the selected locations. SOCAT cells were checked for corresponding SST and salinity values.

GLODAP was tested separately because it was initially intended as an independent DIC source. The QGIS inspection showed that suitable GLODAP observations did not overlap the selected study region. It was therefore excluded rather than using spatially distant observations as substitutes.

This verification stage was important because the different datasets have substantially different spatial resolutions and sampling structures.

4.5 Selection of Meadow and Open-Water Pairs

After dataset coverage was assessed, suitable sampling locations were identified. Each sampling pair consists of:

1. one point located within the mapped seagrass extent, and
2. one nearby point representing surrounding open water.

Eight meadow/open-water pairs were ultimately selected. The paired design provides a direct local comparison while reducing the influence of broad-scale spatial gradients.

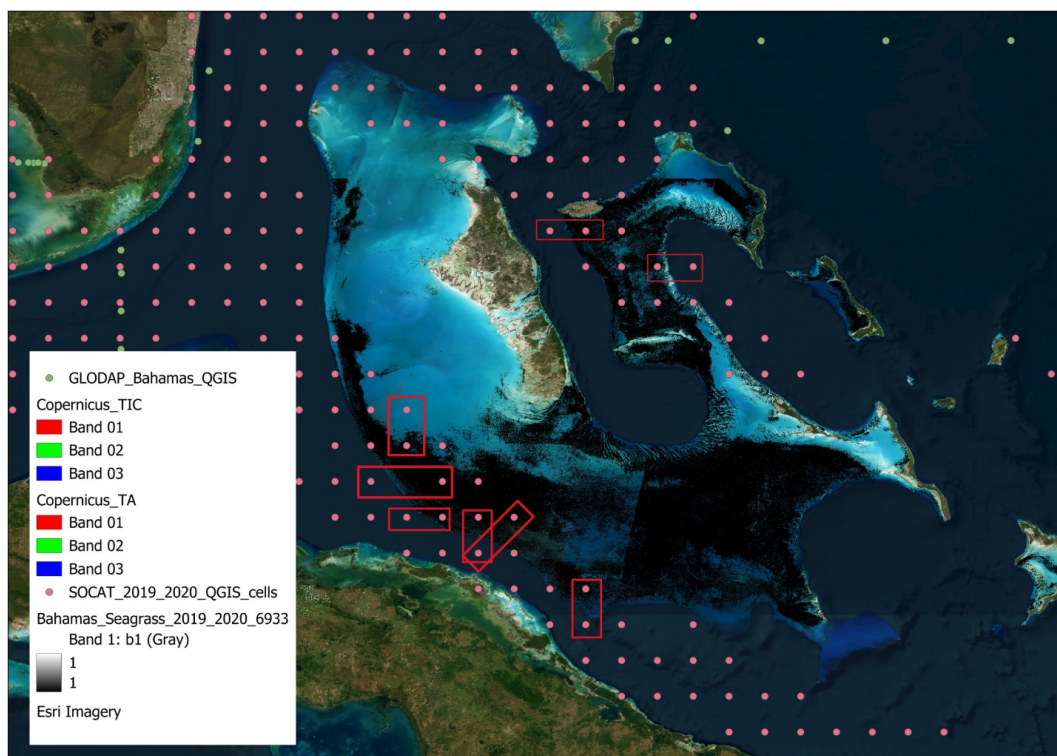


Figure 5: **Eight selected seagrass–open-water sampling pairs over the Bahamas study region.** The spatial positions are shown relative to the mapped seagrass extent and surrounding environmental grids.

Pair	Meadow lat.	Meadow lon.	Open-water lat.	Open-water lon.
1	24.875	−77.375	24.875	−77.625
2	24.625	−76.875	24.625	−76.625
3	23.125	−78.375	23.125	−78.875
4	22.875	−78.375	22.875	−78.625
5	22.875	−78.125	22.625	−78.125
6	22.375	−77.375	22.125	−77.125
7	22.875	−77.875	22.625	−78.125
8	22.375	−77.375	22.125	−77.375

Table 2: The eight meadow/open-water sampling point pairs used in the analysis (degrees; longitudes West).

4.6 Python-Based Data Extraction

Once the spatial locations had been finalised in QGIS, Python scripts were developed to extract and organise the corresponding monthly environmental values.

For each sampling point and month, the workflow extracts:

- DIC/TIC and TA from the Copernicus grid,
- SST and SSS from the SOCATv2023 grid, and
- corresponding Copernicus pH and carbonate variables where available for validation.

The extracted values were stored in tabular form so that the same procedure could be applied consistently across all eight pairs and across the available months.

4.7 PyCO2SYS Carbonate-System Calculation

The extracted DIC and TA values were combined with SST and SSS and supplied to PyCO2SYS. PyCO2SYS solves the marine carbonate system using two carbonate-system parameters together with the required environmental conditions (Humphreys et al., 2022).

For each location,

$$(DIC, TA, SST, SSS) \longrightarrow pH, \Omega_{ar}.$$

The resulting pH and aragonite saturation state were then compared between the meadow and open-water locations.

The differences were defined as

$$\Delta pH = pH_{\text{meadow}} - pH_{\text{open water}},$$

and

$$\Delta \Omega_{ar} = \Omega_{ar, \text{meadow}} - \Omega_{ar, \text{open water}}.$$

A positive ΔpH therefore represents higher calculated pH in the meadow-associated location.

4.8 Monthly Pair Analysis

The PyCO2SYS calculation was performed separately for each available pair and month. Where multiple pairs contained valid data for the same month, their calculated differences were combined to obtain the monthly comparison.

The number of contributing pairs was retained as an important measure of data availability rather than assuming that every monthly result had identical spatial support.

4.9 Observed Carbonate-System Differences

The final comparison showed a consistently positive carbonate-system difference between the seagrass-associated and open-water locations for the common months analysed. The calculated ΔpH values were approximately 0.0135 in April 2019, 0.0169 in May 2019, and 0.0083 in June 2019. Thus, the calculated pH was higher at the seagrass-associated locations in all three months, with the largest difference occurring in May.

A similar pattern was observed for aragonite saturation state. The corresponding $\Delta \Omega_{ar}$ values were approximately 0.087 in April, 0.198 in May, and 0.108 in June. The largest difference again occurred in May, indicating that the strongest contrast in the carbonate system among the analysed months occurred during this period.

Overall, the results indicate a positive seagrass–open-water carbonate-system contrast during the analysed common months. In particular, the simultaneous positive ΔpH and $\Delta \Omega_{ar}$ values suggest that the seagrass-associated locations experienced comparatively higher pH and aragonite saturation state than the paired open-water locations. These results provide the basis for considering the seagrass-associated areas as potential localised pH refugia, while the subsequent validation assesses the reliability of the calculated pH values.

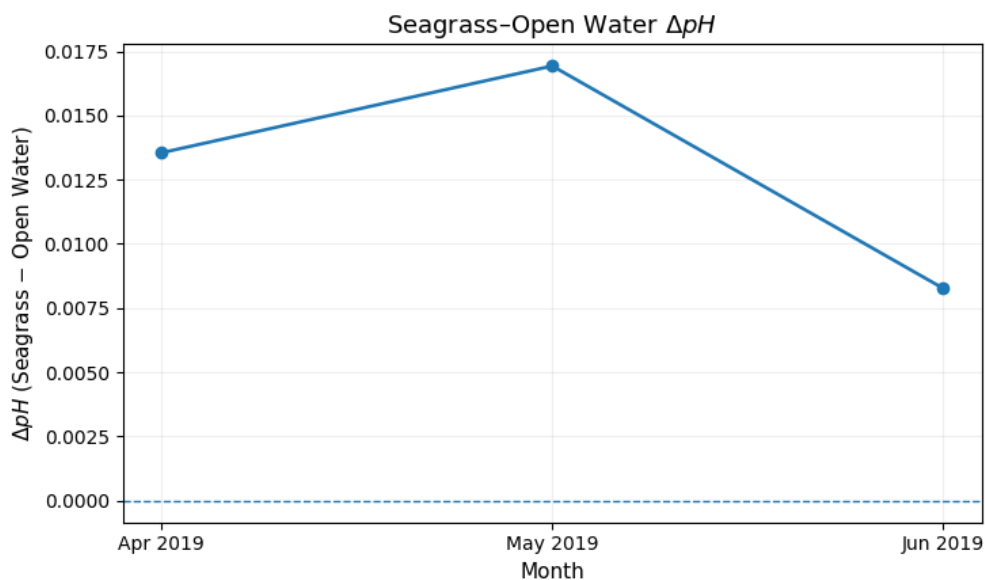


Figure 6: **Seagrass–open-water difference in calculated pH for the available common months.** Positive values indicate higher calculated pH at the seagrass-associated locations. The difference was positive in April, May and June 2019, with the largest ΔpH occurring in May.

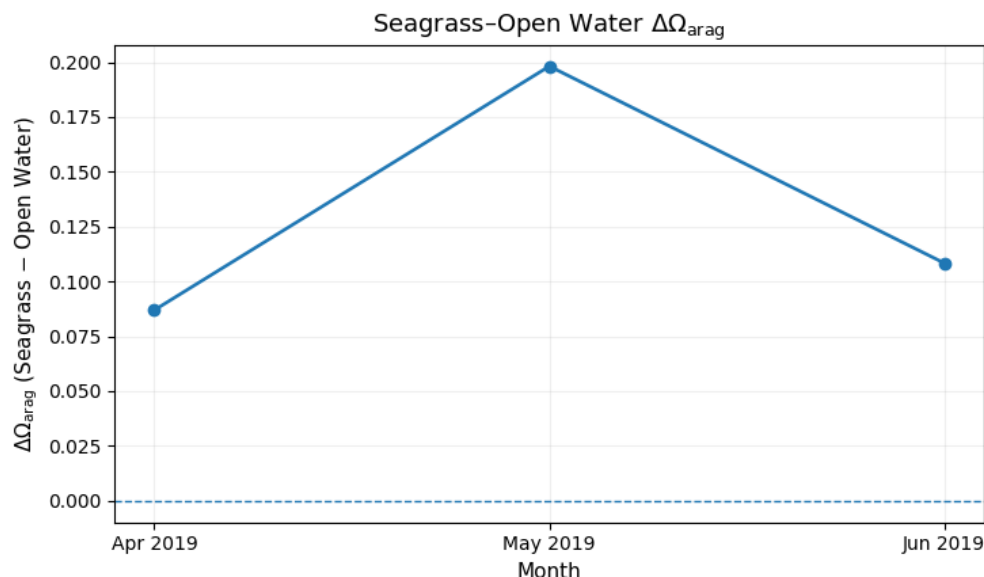


Figure 7: **Seagrass–open-water difference in aragonite saturation state for the available common months.** Positive values indicate higher aragonite saturation state at the seagrass-associated locations. The largest $\Delta\Omega_{ar}$ occurred in May 2019.

4.10 Validation

The calculated PyCO2SYS pH values were compared with the corresponding Copernicus pH values. This provided an independent check of the carbonate-system calculation without using Copernicus pH as an input to PyCO2SYS.

The validation produced an overall mean absolute difference of approximately 0.0104 pH units in the analysed dataset.

4.11 Spatial Reference and Area Handling

The seagrass raster layers were reprojected to EPSG:6933 before area-based calculations. An equal-area projection was used so that spatial area estimates were not directly calculated in geographic latitude/longitude coordinates.

5 Applications

- **Marine spatial planning:** identifying Bahamian seagrass banks with favourable local carbonate conditions can inform coastal zone management and marine protected area design.
- **Reef and calcifier conservation:** carbonate-system information can help identify areas where seagrass habitat and calcifying ecosystems occur in close proximity.
- **Climate adaptation planning:** mapped carbonate variability can contribute to regional assessments of vulnerability to ocean acidification.
- **Seagrass restoration prioritisation:** spatial information on carbonate conditions may provide an additional criterion when evaluating restoration sites.

6 Future Scope

- **More sampling pairs, more consistently populated.** Expanding the number of suitable pairs would improve spatial representation and reduce dependence on a small number of locations.
- **Revisit Google Earth Engine at scale.** With greater computational resources, the initial GEE workflow could be extended to a larger portion of the Bahama Bank.
- **Independent field validation.** In-situ pH and carbonate-system measurements at selected locations would provide direct validation of the gridded-data approach.
- **Biological outcome linkage.** Pairing carbonate-system estimates with calcifier growth or survival measurements would allow the chemical signal to be connected to ecological outcomes.
- **Multi-year extension.** Extending the analysis beyond 2019–2020 would allow interannual variability to be investigated.
- **Alternative DIC source.** Since GLODAP was unavailable for the study area, alternative regional or model-based DIC products could be evaluated against Copernicus.

7 Limitations and Data Credibility Assessment

- The 10-m seagrass map has substantially higher spatial resolution than the Copernicus carbonate-system fields. Consequently, a Copernicus grid cell can contain both seagrass and open-water environments.
- The eight-point-pair design represents selected locations rather than a continuous map of the entire seagrass system.
- Copernicus TA/DIC fields are gridded oceanographic products rather than direct measurements at each selected point.
- SOCATv2023 provides observation-based gridded environmental information, but its sampling density varies spatially and temporally.
- GLODAP was excluded because suitable observations were not available in the study area rather than using spatially distant observations as substitutes.
- The 2019–2020 study period provides limited temporal coverage for assessing long-term trends.
- The calculated carbonate differences indicate a potential chemical signal; they do not independently demonstrate an ecological refugium effect.

8 Conclusion

This project develops a reproducible workflow for investigating potential seagrass-associated carbonate-system differences in the Bahamas using multiple complementary datasets. The workflow begins with high-resolution seagrass mapping and spatial processing, proceeds through QGIS-based dataset verification and sampling-point selection, and then uses Python to extract monthly environmental data and PyCO2SYS to calculate pH and aragonite saturation state.

The combination of datasets is important because no individual source provides all of the information required. The 10-m seagrass product identifies the habitat, Copernicus provides TA and DIC, SOCATv2023 provides SST and SSS, and PyCO2SYS combines these inputs into a consistent carbonate-system calculation. GLODAP was explicitly tested but excluded because suitable coverage was unavailable within the study area.

The final ΔpH and $\Delta\Omega_{ar}$ comparisons provide the principal outputs of the workflow. The major methodological limitation remains the mismatch between the fine spatial resolution of the seagrass mapping and the much coarser carbonate-system grids. Future spatial downscaling and independent field measurements can therefore improve the ability of the workflow to resolve meadow-scale carbonate variability.

References

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